

PROFESSIONAL SUMMARY

Results-driven Full-Stack Software Engineer with hands-on experience building scalable marketplace applications using modern technologies. Proficient in **Node.js**, **React Native (Expo)**, **TypeScript**, and **Azure Cloud Services** with expertise in **RESTful API design**, **microservices architecture**, and **containerized deployments** using Docker and Kubernetes.

Demonstrated success developing end-to-end e-commerce solutions featuring **real-time payment processing (Stripe)**, **third-party API integrations (FedEx, OAuth 2.0)**, and **event-driven architectures** with Azure Service Bus and Azure Functions. Strong foundation in **database design** and **query optimization** across PostgreSQL and Azure Cosmos DB, with caching strategies that reduced operational costs by 70%.

Azure Certified Developer Associate with experience deploying **serverless functions**, managing cloud infrastructure on **Azure App Services** and **AKS**, and implementing **CI/CD pipelines**. Skilled at building **mobile-first applications**, writing comprehensive test suites with **Jest**, and following security best practices including **JWT authentication** and input validation.

Passionate about writing clean, maintainable code and collaborating in **Agile/Scrum environments** to deliver high-quality software solutions.

Profile Links:

- GitHub: <https://github.com/ajay99511>
- LinkedIn: <https://www.linkedin.com/in/e-aj-47b71238b/>
- LeetCode: <https://leetcode.com/u/ajay216/>

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, C#, Java, SQL, PL/SQL

Front-End: React.js, React Native, Angular, Next.js, Redux, HTML5, CSS3, Bootstrap, jQuery

Back-End: Node.js, Express.js, ASP.NET Core, ASP.NET MVC, .NET Core, Flask

Databases: PostgreSQL, SQL Server, MySQL, Azure Cosmos DB, RDS, Redis (caching)

Cloud & DevOps: Azure (App Service, Functions, Blob Storage, Service Bus, Key Vault, AKS), AWS, Docker, Kubernetes, CI/CD, Azure DevOps

APIs & Integration: RESTful APIs, GraphQL, Stripe, FedEx API, OAuth 2.0, Socket.io, Passport.js

Testing & Tools: Jest, NUnit, Test-Driven Development (TDD), Git, GitHub, TFS, Visual Studio Code

Development Practices: Agile/Scrum, Microservices Architecture, Event-Driven Architecture, SDLC

Additional Tools: BullMQ, Multer, Axios, Zod, Pino, RabbitMQ, Jupyter Notebook

WORK EXPERIENCE

Software Engineer | SLK Holdings | Jul 2024 - Present

- Developed full-stack cloud-native applications using Node.js (Express.js), React.js, and React Native (Expo) to handle real-time interactions and transactional workflows.
- Implemented RESTful APIs using Express.js routers to handle user authentication, resource management, and asynchronous workflow orchestration.
- Created responsive web user interfaces using React.js and configured Redux Toolkit for predictable global state management across complex data flows.
- Developed cross-platform mobile applications using React Native, TypeScript, and Expo Router to deliver native performance and tab-based navigation.
- Connected Redux asynchronous thunks with Express backend endpoints to handle data fetching, loading states, and smooth UI updates.
- Set up a microservices system to isolate high-volume webhook handling, third-party integrations, and automated background jobs.
- Deployed and managed serverless workloads and containerized microservices using Azure App Services, Azure Functions, and Azure Blob Storage.
- Implemented event-driven architectures using Azure Service Bus for asynchronous message queuing and reliable background task coordination.
- Containerized application services using Docker and managed orchestration across cloud environments using Kubernetes on Azure Kubernetes Service (AKS).
- Created normalized PostgreSQL database schemas and implemented database access layers using Sequelize ORM to abstract data operations.
- Integrated Azure Cosmos DB for flexible, document-based storage of user activity logs, session events, and

semi-structured analytics data.

- Optimized SQL database performance through strategic indexing, query execution planning, and connection pooling configurations using PgBouncer.
- Configured high-performance caching layers using Redis to store frequently accessed API responses, authentication tokens, and session states.
- Implemented bidirectional, real-time messaging features using Socket.io with message persistence and delivery confirmations.
- Integrated secure payment processing workflows supporting payment intents, webhooks, and multi-step financial transaction flows with error recovery.
- Developed asynchronous background workers and task queues using BullMQ and Redis to process notifications, cleanup jobs, and scheduled reports.
- Configured OAuth 2.0 authentication strategies using Passport.js to support secure Google, Apple, and Facebook single sign-on.
- Applied custom Express middleware and validation libraries like Zod to enforce strict input validation, sanitize requests, and prevent injection vulnerabilities.
- Managed sensitive environment configurations and application credentials securely using Azure Key Vault across staging and production environments.
- Wrote global error handling middleware, custom exception classes, and centralized application logging infrastructure using Winston.
- Developed reusable UI components and interactive forms using React Hook Form, incorporating client-side validation and layout primitives.
- Wrote comprehensive unit and integration test suites using Jest and Supertest to validate API endpoints, middleware logic, and business services.
- Implemented isolated database state strategies using transactional test fixtures to ensure testing data integrity and automatic rollback capabilities.
- Executed load testing scenarios and performance profiling to isolate bottlenecks in database queries, Express middleware execution, and component rendering.
- Configured continuous integration and deployment pipelines to automate testing, Docker image building, and deployment updates across cloud infrastructure.

GRADUATE PROJECTS

Full-Stack Dating Application | University of Central Missouri

Advanced Database Systems (CS 5600) | Sep 2023 - May 2024

GitHub: <https://github.com/ajay99511/DatingApp216>

Tech Stack: .NET 8, Angular 16, SQL Server 2022, Entity Framework, SignalR, ADO.NET

- Architected and developed full stack dating application with real-time messaging capabilities serving **100+ concurrent users**
- Designed normalized **SQL Server database** with Entity Framework, implementing data access layer optimizing relational database interactions and query performance
- Built **real-time communication features** using **SignalR**, enabling instant messaging and push notifications between users with **<500ms latency**
- Developed **RESTful API endpoints** in C# following industry best practices for secure front-end-backend communication
- Optimized database performance by writing **stored procedures** handling complex queries, reducing response time by **35%** under heavy loads
- Enhanced security implementing **Role-Based Access Control (RBAC)** using Identity Framework with JWT authentication
- Created modular **Angular frontend** with reusable components, services, route guards, and interceptors ensuring secure and responsive UI
- Configured deployment and hosting on **IIS (Internet Information Services)** via port 8081, simulating production environment

Key Achievement: Successfully delivered end-to-end dating platform demonstrating proficiency across full technology stack

Diabetes Prediction Using Machine Learning | University of Central Missouri

Machine Learning (CS 5710) | Jan 2023 - May 2023

Tech Stack: Python, scikit-learn, Flask, Pandas, NumPy, Seaborn, Matplotlib

- Developed diabetes prediction model utilizing **ensemble learning** techniques (SVM, KNN, Random Forest) achieving **87% accuracy**

- Automated **data preprocessing pipeline** with Python and scikit-learn, including feature engineering (BMI-to-Age ratio), standardization, and handling missing values for **1,000+ patient records**
- Applied feature selection and **correlation analysis** using heatmap visualization to identify key relationships between 8 health indicators
- Visualized dataset distributions with **Seaborn** bar and pie charts, improving model interpretability and stakeholder presentations
- Deployed **REST API using Flask** delivering real-time predictions with **<200ms response time**, integrating model storage with local file systems
- Evaluated model performance through **cross-validation**, with Random Forest outperforming baseline by **12% accuracy**
- Analyzed **feature importance** providing actionable insights into key factors influencing diabetes risk (glucose levels, BMI, age)

Key Achievement: Delivered production-ready ML model with API integration demonstrating end-to-end data science capabilities

Soil Moisture Prediction Using Random Forest | Undergraduate Final Project

Vasireddy Venkatadri Institute of Technology, India | May 2021 - Mar 2022

Tech Stack: Python, scikit-learn, Pandas, NumPy, SciPy, Matplotlib

- Built **Random Forest Classifier** predicting groundwater availability with **82% accuracy** using environmental and geographical features
- Extracted and engineered relevant features including usage-to-availability ratio to enhance model performance and interpretability
- Performed comprehensive **data preprocessing**: removed null values, handled missing data, and normalized features across **5,000+ data points**
- Applied statistical and numerical techniques using **SciPy** and Pandas to clean and transform raw data for efficient ML modeling
- Implemented **feature scaling** and transformation techniques, standardizing input features to improve algorithm convergence
- Utilized scikit-learn to train and tune Random Forest Classifier with **hyperparameter optimization**, achieving robust predictive model

Key Achievement: Demonstrated ability to apply ML techniques to real-world environmental challenges with practical agricultural applications

EDUCATION

University of Central Missouri, Warrensburg, MO | GPA 3.5

Master of Science in Computer Science | Dec 2022 - May 2024

Relevant Coursework: Advanced Database Systems, Machine Learning, Software Engineering, Cloud Computing

Jawaharlal Nehru Institute of Technology, Hyderabad, India | GPA 3.13

Bachelor of Technology in Computer Science | Aug 2018 - May 2022

CERTIFICATIONS

- **Microsoft Certified: Azure Developer Associate (AZ-204)** | Microsoft | 2024
- **Technical Support Fundamentals** | Google (via Coursera) | 2023
- **Python for Everybody Specialization** | University of Michigan (via Coursera) | 2022

ADDITIONAL INFORMATION

- **Hackathon Experience:** Active participant in coding competitions to enhance problem-solving skills and build portfolio projects
- **Continuous Learning:** Regularly exploring AI-assisted development tools and modern app development frameworks
- **Open Source:** Actively maintaining GitHub repositories showcasing full-stack projects and technical documentation